6 Damped Harmonic Oscillator Lab Guide

Introduction

The harmonic oscillation experimental system is described in figure 6.1: Two ideal springs with elastic constants k_1 and k_2 are connected to the ends of a cart that is placed on a level frictionless track. The other ends of the springs are connected to the ends of the track.

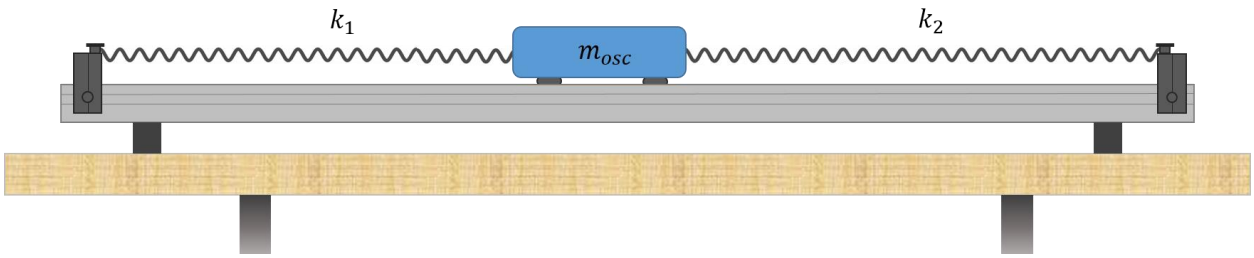


Figure 6.1 Harmonic oscillation experimental setup

At equilibrium, the total force acting on the cart equals to zero. If the cart is displaced distance x_{osc} from equilibrium, the springs will exert on it a force that will try to restore the cart to its equilibrium position. This force is proportional to the amount of displacement x_{osc} and its total value will be given by Hooke's law:

$$F_{spring} = -(k_1 + k_2)x_{osc} \qquad \text{(eq. 1)}$$

The minus sign imply that the force is directed in the opposite direction of the displacement, a restoring force.

Assuming negligible friction, once the cart is released, one spring will push the cart, and the other spring will pull the cart while the cart displaces and acquire momentum as the spring's potential energy turns into kinetic energy. And the cycle will continue indefinitely, and the cart moves harmonically. However, if some friction (or drag) between the cart and the track exists, then energy will dissipate from the system in form of heat, and the amplitude of the motion of the cart will decrease until the cart stops.

<u>A question arises:</u>

If friction, or drag, exists in a harmonically moving system, how fast will the amplitude decrease? Will the amplitude decrease linearly? Or maybe in a different way (for example, maybe exponentially)? In today's experiment, we will use curve fitting to determine the cause of friction in the oscillator system and we will find the magnitude of the friction coefficient.

Theoretical model

The equation of motion for the oscillating cart on a frictionless levelled track is

$$m_{osc}\frac{d^2x_{osc}}{dt^2} = -(k_1 + k_2)x_{osc} \qquad \text{(eq. 2)}$$

Let us define the natural angular frequency of the oscillator, ω_0,

$$\omega_0 = \sqrt{\frac{k_1 + k_2}{m_{osc}}} \qquad \text{(eq. 3)}$$

Thus, we rewrite the equation of motion into standard form as,

$$\ddot{x}_{osc} = -\omega_0^2 x_{osc} \qquad \text{(eq. 4)}$$

The solution of this differential equation, $x_{osc}(t)$ describes the position of the oscillating cart as function of time. We are looking for a function whose 2nd derivative is proportional to the negative of itself. We are familiar with some functions with this property, namely sines, cosines, and exponentials. Let us try,

$$x_{osc}(t) = A\cos(\omega_0 t + \phi) \qquad \text{(eq. 5)}$$

This equation oscillation with angular frequency ω_0, amplitude A, and phase ϕ (figure 6.2, left).

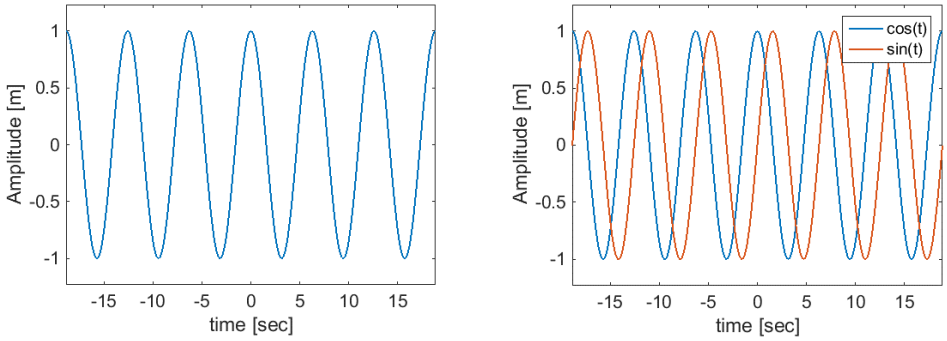


Figure 6.2 (left) Cosine. (right) Cosine and a Sine.

Note: A sine or an exponential function with complex argument would work just as well. But a sine function is simply a shifted cosine function by a $\pi/2$ phase, so it doesn't really generate anything new; it just changes the phase (figure 6.2, right).

The *intensity* of the wave is proportional to the square of the *amplitude*, A $[m]$. *Frequency, ω,* is the rate, or number of times per second, that a wave cycles from positive to negative to positive again. Frequency is measured in cycles per second or hertz (Hz). For sound waves, the human ear interprets amplitude as volume or loudness and frequency as the pitch of sound. The *phase* of a wave, ϕ, is related to the displacement at $t = 0$.

In our experiment, we shift the cart at time $t = 0$ to a distance A_0 from the equilibrium point, therefore the equation describing the motion of the cart shall be:

$$\boldsymbol{x_{osc}(t) = A_0\cos(\omega_0 t)} \qquad \textbf{(eq. 6)}$$

Notice that, for any initial conditions, the cart will oscillate with ω_0 frequency determined by the physical properties of the system: the spring constant k and the oscillating mass m. For this reason, ω_0, is called the natural frequency of the oscillator.

Energy of a simple harmonic oscillator

Assuming no friction, the restoring force of a spring, $F = -kx$, is a conservative force. Therefore, the energy of the harmonic oscillator is conserved. The energy is the sum of kinetic energy, $K = \frac{1}{2}m_{osc}\dot{x}_{osc}^2$, and the potential energy, $U = \frac{1}{2}kx_{osc}^2$. The energy of a harmonic wave is,

$$E_{tot} = \frac{1}{2}m_{osc}\dot{x}_{osc}^2 + \frac{1}{2}kx_{osc}^2 = \frac{1}{2}m_{osc}A^2\omega_0^2 \qquad \text{(eq. 7)}$$

Figure 6.3 shows how the energy breaks up into kinetic and potential, as a function of time (Note that we have arbitrarily chosen zero phase, $\phi = 0$).

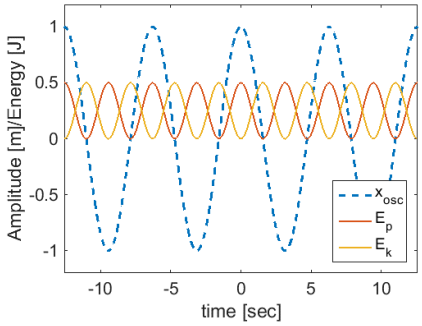


Figure 6.3 Energy of a harmonic oscillator

The energy oscillates back and forth between kinetic and potential. It is all potential at the points of maximum stretching, and it is all kinetic when the mass passes through the origin. However, total energy is constant at any moment since we have assumed frictionless model.

Damped Harmonic Oscillations

In the absence of any form of friction, the system will continue to oscillate with no decrease in amplitude. However, if there is some form of friction, then the amplitude will decrease as a function of time since energy will be lost in form of heat (figure 6.4). Such friction can be result of kinetic friction or drag force.

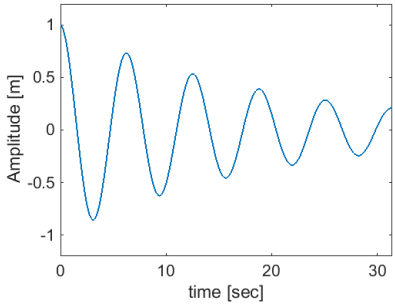


Figure 6.4 Damped Harmonic Oscillations

Harmonic motion with kinetic friction

Kinetic friction (also called sliding friction) occurs when two objects in contact are moving relative to each other. Kinetic friction is a constant force proportional to the normal force, $N = mg$, where the proportion coefficient is the kinetic friction coefficient, μ_k.

$$f_k = -\mu_k N \qquad \text{(eq. 8)}$$

The minus sign stands for the fact that the friction force acts in the opposite direction to body's motion. The friction coefficient, μ_k, is a physical parameter depending on the material type and shape of each of the sliding objects.

Suppose we displace the mass by distance A_0 from equilibrium, the system has potential energy $U_0 = \frac{1}{2}kA_0^2$. Then we release the mass and wait until it reaches maximum displacement, A_1 (where $A_1 < A_0$) the energy of the system at this point is $U_1 = \frac{1}{2}kA_1^2$. Since the kinetic friction force is a constant force, the work done by the friction is a constant, times the distance travelled $(A_0 + A_1)$.

$$U_0 - U_1 = (A_0 + A_1)|f_k| \qquad \text{(eq. 9)}$$

That is,

$$\frac{1}{2}kA_0^2 - \frac{1}{2}kA_1^2 = (A_0 + A_1)\mu_k m_{osc} g \qquad \text{(eq. 10)}$$

Solving for A_1, we find

$$\boldsymbol{A_1 = A_0 - \frac{2|f_k|}{k} = A_0 - \frac{2\mu m_{osc} g}{k}} \qquad \text{(eq. 11)}$$

Thus, at each turning point (when the mass reaches maximum displacement), the amplitude (and the energy) decreases by a constant amount, $2\mu m_{osc} g/k$, compared to the previous turning point. Hence, the amplitude (and the energy) of the oscillation will decrease *linearly* with time until the oscillation stops completely.

More details regarding harmonic motion with kinetic friction force can be found in references 3 and 4.

Harmonic motion with drag force

Drag force is a type of frictional force which is proportional to the body's velocity,

$$f_d = -b\dot{x}_{osc} \qquad \text{(eq. 12)}$$

Where b is a constant describing the strength of damping.

In the case when drag force is present, the equation of motion of the oscillating cart is given by,

$$m_{osc}\frac{d^2x_{osc}}{dt^2} = -(k_1 + k_2)x_{osc} - b\dot{x}_{osc} \qquad \text{(eq. 13)}$$

Then, the equation of motion of the oscillator can be rewritten as follows:

$$\ddot{x}_{osc} + 2\gamma\dot{x}_{osc} + \omega_0^2 x_{osc} = 0 \qquad \text{(eq. 14)}$$

Where $\gamma = b/2m_{osc}$ and $\omega_0^2 = k/m_{osc}$ as before.

For *lightly damped* oscillator, $\omega_0 \gg \gamma$, the solution of the equation of motion gives the position of the damped oscillator at any time,

$$x_{osc}(t) = Ae^{-\gamma t}\cos(\omega_1 t + \phi_1) \quad \textbf{(eq. 15)}$$

where we have defined the proper frequency of the oscillator, ω_1, as

$$\omega_1 = \sqrt{\omega_0^2 - \gamma^2} \quad \textbf{(eq. 16)}$$

Hence, in the case of damping due to drag force, the amplitude of the oscillator decays *exponentially* with characteristic time, $\tau = 1/\gamma$.

Note: The proper frequency can be a complex number. The solution presented here, and the one that we will discuss in the laboratory is the case of underdamping. Two more cases are possible: Overdamping, $\omega_0 < \gamma$, then ω_1 is purely imaginary number and the last case is called critical damping, for the case where $\omega_0 = \gamma$, then $\omega_1 = 0$. For the cases of overdamping and underdamping the motion will be exponentially decaying without oscillations.

In our experiment we produce drag force by magnetic braking. Magnetic braking is achieved by attaching magnets to the end of the cart (as we saw in the terminal velocity experiment).

Experimental procedure

a. Verify that the track is stable and levelled.

b. Using the electronic scale weigh the mass of the cart, the magnetic damping device and one of the extra masse blocks and record the values including the measurement errors (uncertainties)

c. Place one of the extra masse blocks on top of the cart.

d. **The magnetic damping device must be attached all the time to the cart, because we don't want to change the mass of the oscillator (hence, the natural frequency) during the experiment. In experiment part (f) you have to attach the magnets to the cart in such a way that they do not produce magnetic damping. In experiment part (g), you will place the magnets in such way they will produce magnetic damping to the motion of the cart.**

e. Measure the spring constant of the two-spring system, k_1 and k_2, according to the procedure explained to you by the laboratory tutor.

f. In PASCO Capstone, click on 'file' and open an experiment named 'HO exp setup.cap' that you have downloaded from Moodle.

g. <u>Harmonic oscillations experiment:</u>

f.1 Setup the experimental system as it described in figure 6.1. Make sure that the spring are not stretched beyond their linear length!

f.2 Tap the cart a few times so that it moves to its equilibrium position.

f.3 Start recording the car position.

f.4 Displace the cart from equilibrium position (it is advised **not** to displace to cart more than 10 cm). Next, release the cart from rest. This situation is equivalent to initial conditions:

$$\begin{cases} x_{osc}(0) = A \\ \dot{x}_{osc}(0) = 0 \end{cases}$$

Continue recording the cart's displacement until the oscillations decay completely.

h. Magnetically damped harmonic oscillations

g.1 Place the Magnetic damping device on the magnet-equipped end of the cart when the magnets are facing the track.

g.2 Slide the Magnetic damping device up or down to adjust the amount of damping. The magnets should be parallel to the track. They should not touch the track but should be close enough in order to induce sufficient magnetic damping.

g.3 Start recording the cart position. Displace the cart distance x_{max} from equilibrium and release it from rest. **A good measurement will have about 6-8 oscillations until the cart stops moving.**

Data analysis

1. Open a new python script and write down the experiment log: Write in all the directly measured parameters and their uncertainties. Using comments, include the units of every measured parameter.
2. **Calculate the values and uncertainties of the spring constants k_1 and k_2 of your springs and the natural frequency of the system ω_0.**
3. **Import the data from PASCO capstone as explained in the Appendix at the end of this guide.**
4. Harmonic oscillations experiment (experiment part f):
   - **4.1 Plot graph of the position of the cart is function of time, $x_{osc}(t)$. Verify that the graph represents the solution of the equation of motion with the initial conditions of the experiment, you may need to remove data points and rearrange the data to achieve this.**
   - 4.2 Use Python function *scipy.signal.find_peaks* to find the turning points of the cart and add these point to the plot of $x_{osc}(t)$.
   - **4.3 If you find that the harmonic oscillations are damped (if the amplitude of the oscillations is decreasing), use curve fitting and goodness of fit (r^2 and residuals are needed, read chapter 5.1 curve fitting) to determine whether the damping is due to sliding friction or drag force. Add the fit line to the plot of $x_{osc}(t)$.**
   - 4.4 Find the damping strength (friction coefficient, μ_k, for the case of friction or b for the case of drag).
   - **4.5 Find the experimental natural frequency of the oscillator, ω_0, and compare your result to the theoretical value. What is your conclusion from this comparison?**
5. Magnetically damped harmonic oscillations (experiment part g):
   - **5.1** Plot graph of the position of the cart is function of time, $x_{osc}(t)$. **Verify that the graph represents the solution of the equation of motion with the initial conditions of the experiment, you may need to remove data points and rearrange the data to achieve this.**
   - 5.2 Use Python function *scipy.signal.find_peaks* to find the turning points of the cart and add these point to the plot of $x_{osc}(t)$.
   - **5.3 Use curve fitting and goodness of fit (r^2 and residuals are needed) to determine whether the damping is due to sliding friction or drag force. Add the fit line to the plot of $x_{osc}(t)$.**
   - 5.4 Find the damping strength (friction coefficient, μ, for the case of friction or τ for the case of drag).
   - **5.5 Find the experimental natural frequency of the oscillator, ω_1, and compare your result to the theoretical value. What is your conclusion from the comparison?**
6. **At the end of work, upload your work files (data file and Python script) to the course's site under the damped harmonics oscillator section. Without these files there will not be grade for the lab work!**

References

[1] Walker, J., Halliday, D., Resnick, R., & Trees, B. R. (2018). *Fundamentals of physics 9th edition*. Ch. 15, pp. 385-412 Wiley.

[2] Images of system parts were taken from PASCO site: www.pasco.com

[3] Kamela, M. (2007). An oscillating system with sliding friction. The Physics Teacher, 45(2), 110-113.

[4] Marchewka, A., Abbott, D. S., & Beichner, R. J. (2004). Oscillator damped by a constant-magnitude friction force. American journal of physics, 72(4), 477-483.

[5] By Chetvorno - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=96468501